MRC Lecture 5 — Derivation Guide 2

The Position Weight Matrix as a Dot Product

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1. Motivation

This is the **central derivation of Lecture 5** and one of the most important in the course. Position weight matrices (PWMs) have been the field-standard representation of transcription factor (TF) binding specificity since Staden (1984) and Stormo (1986) — decades before the transformer architecture was introduced. A PWM is typically presented as a 4×L matrix of log-odds scores, and "scoring a site" is presented as looking up the appropriate log-odds at each position and summing. This guide shows that **the PWM scoring operation is mathematically a dot product between the TF's feature vector and the DNA's feature vector**. Not an analogy. Not an approximation. The same operation that transformers perform in their attention mechanism. The biology community has been computing attention scores — the core operation of modern AI — since 1984 under a different name.

Because the PWM scoring IS the transformer attention score, TF-DNA binding is the cleanest possible demonstration of the **convergence equation**: the attention equation derived from engineering and the Boltzmann selection probability for TF binding are the same equation, and the PWM formalism is where this is most transparent.

2. The PWM Formalism

Setup. A transcription factor has a characteristic binding motif of length L. The position weight matrix W is a real-valued matrix of dimension $L \times 4$:

$$W = \begin{pmatrix} W_{1,A} & W_{1,C} & W_{1,G} & W_{1,T} \\ W_{2,A} & W_{2,C} & W_{2,G} & W_{2,T} \\ \vdots & \vdots & \vdots & \vdots \\ W_{L,A} & W_{L,C} & W_{L,G} & W_{L,T} \end{pmatrix}$$

Each entry $W_{p,b}$ is the log-odds contribution of nucleotide $b \in \{A, C, G, T\}$ at position p:

$$W_{p,b} = \ln \frac{P(b \mid \text{bound position } p)}{P(b \mid \text{background})}$$

where $P(b \mid \text{bound position } p)$ is estimated from aligned binding sites (e.g., from ChIP-seq or SELEX experiments) and $P(b \mid \text{background})$ is the nucleotide frequency in non-bound DNA (typically 0.25 for each base). Positive $W_{p,b}$ values indicate that base b is enriched at position p in bound sites; negative values indicate depletion.

Scoring a candidate DNA site. Given a candidate DNA sequence $d = d_1 d_2 \dots d_L$, the PWM score is computed by summing the log-odds contributions at each position:

$$S(d; W) = \sum_{p=1}^{L} W_{p,d_p}$$

This score is proportional to the log-probability that the TF binds d versus the background. Higher score = more TF preference. This is the scoring function used in FIMO, HOMER, JASPAR, Cis-BP, and virtually every TF binding-site prediction tool in computational genomics.

3. The DNA Sequence as a One-Hot Vector

Represent the DNA sequence d as a one-hot encoded matrix $X \in \{0,1\}^{L\times 4}$:

$$X_{p,b} = \begin{cases} 1 & \text{if } d_p = b \\ 0 & \text{otherwise} \end{cases}$$

For example, the DNA sequence ACGT is encoded as:

$$X_{\text{ACGT}} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

(rows are positions 1–4, columns are A, C, G, T)
Exactly one entry is 1 in each row; the rest are 0. This is the canonical representation for categorical features in machine learning.

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4. The PWM Score as a Dot Product

We can now rewrite the PWM scoring equation as a dot product.

Form 1: Double sum.

The original scoring equation:

$$S(d;W) = \sum_{p=1}^{L} W_{p,d_p}$$

Using the one-hot encoding, $W_{p,d_p} = \sum_b W_{p,b} X_{p,b}$ (exactly one term in the inner sum is non-zero).
Therefore:

$$S(d;W) = \sum_{p=1}^{L} \sum_{b \in \{A,C,G,T\}} W_{p,b} X_{p,b}$$

This is the **element-wise product of W and X, summed over all entries**. In matrix notation:

$$S(d;W) = \mathrm{tr}(W^T X)$$

or equivalently, the Frobenius inner product $\langle W, X \rangle_F$.

Form 2: Flattened dot product.

Flatten W into a $4L$-dimensional vector:

$$w = (W_{1,A}, W_{1,C}, W_{1,G}, W_{1,T}, W_{2,A}, \dots, W_{L,T})$$

Flatten X similarly:

$$x = (X_{1,A}, X_{1,C}, X_{1,G}, X_{1,T}, X_{2,A}, \dots, X_{L,T})$$

Then:

$$S(d;W) = w \cdot x$$

The PWM score is a dot product between two $4L$-dimensional vectors. The first vector (w) encodes the TF's binding preferences (the flattened PWM). The second vector (x) encodes the DNA sequence (the flattened one-hot encoding). This is the same mathematical object as transformer attention — one vector from the "query" (TF preferences), one from the "key" (DNA features), producing a scalar compatibility score.

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5. Worked Example: A Four-Position TF Motif

The TF. A hypothetical transcription factor with a strong preference for the motif ACGT. Its PWM:

$$W = \begin{pmatrix} 2 & 0 & -1 & -1 \\ -1 & 2 & 0 & -1 \\ -1 & 0 & 2 & -1 \\ -1 & -1 & 0 & 2 \end{pmatrix}$$

(rows: positions 1–4; columns: A, C, G, T)
Reading the diagonal: A at position 1 scores +2, C at position 2 scores +2, G at position 3 scores +2, T at position 4 scores +2. All other nucleotides at these positions score either 0 or −1. The motif ACGT gives the maximum score.

Scoring a matching site (ACGT):

$$X_{\text{ACGT}} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

Using the double-sum form:

$$S = \sum_p \sum_b W_{p,b} X_{p,b} = W_{1,A} + W_{2,C} + W_{3,G} + W_{4,T} = 2+2+2+2 = 8$$

Alternatively, as a flattened dot product. The flattened PWM:

$$w = (2,0,-1,-1,-1,2,0,-1,-1,0,2,-1,-1,-1,0,2)$$

The flattened DNA:

$$x = (1,0,0,0,0,1,0,0,0,0,1,0,0,0,0,1)$$

Dot product:

$$w \cdot x = 2(1) + 0(0) + (-1)(0) + (-1)(0) + (-1)(0) + 2(1) + \ldots + 2(1) = 2+2+2+2 = 8 \checkmark$$

Same answer — now visibly a dot product.

Scoring a mismatching site (TGCA):

The DNA has all the “wrong” nucleotides at every position:

$$S = W_{1,T} + W_{2,G} + W_{3,C} + W_{4,A} = (-1) + 0 + 0 + (-1) = -2$$

The mismatching site scores 10 points lower than the matching site.

Scoring a partial match (ACCT):

Positions 1, 2 match; position 3 has C instead of G; position 4 matches:

$$S = W_{1,A} + W_{2,C} + W_{3,C} + W_{4,T} = 2+2+0+2 = 6$$

Intermediate score — two points lower than perfect match.

6. The Convergence Equation for TF-DNA Binding

Now apply the Boltzmann distribution to competitive TF binding. The TF encounters three candidate DNA sites with scores:

- Site 1: ACGT → $S_1 = 8$
- Site 2: ACCT → $S_2 = 6$
- Site 3: TGCA → $S_3 = -2$

Boltzmann selection probability (with $\tau = 1$ for simplicity):

$$P(\text{TF binds site } j) = \frac{\exp(S_j)}{\sum_k \exp(S_k)}$$

Step 1: Exponentiate.

$$\exp(8) = 2981$$

$$\exp(6) = 403$$
$$\exp(-2) = 0.135$$

Step 2: Partition function.

$$Z = 2981 + 403 + 0.135 = 3384$$

Step 3: Probabilities.

$$P_1 = 2981 / 3384 = 0.881$$
$$P_2 = 403 / 3384 = 0.119$$
$$P_3 = 0.135 / 3384 = 0.0000399$$

Interpretation. The perfect-match site captures 88% of the binding probability. The partial-match site gets 12%. The non-match site is essentially invisible. This is exactly how TF binding works in real biology — high-affinity consensus sites dominate, with measurable but minor binding to suboptimal sites.

7. The Identity with Transformer Attention

Writing the two equations side by side:

PWM Boltzmann selection (since ~1984):

$$P(\text{bind site } j) = \frac{\exp(w_{\text{TF}} \cdot x_j / \tau)}{\sum_k \exp(w_{\text{TF}} \cdot x_k / \tau)}$$

Transformer attention (since 2017):

$$A_j = \frac{\exp(q \cdot k_j / \tau)}{\sum_k \exp(q \cdot k_k / \tau)}$$

Under the identification:

- $w_{\text{TF}} \leftrightarrow q$ (the TF's preference vector is the query)
- $x_j \leftrightarrow k_j$ (the DNA's one-hot encoding is the key)
- $\tau \leftrightarrow \tau$ (the temperature — physical RT in biology, $\sqrt{d_k}$ in transformers)

These are the same equation. The community of TF biophysicists has been performing transformer attention since 1984, under a different name and for different reasons. Stormo (2000) derived the PWM framework from information theory; Vaswani et al. (2017) derived attention from dot-product similarity in machine translation. They converged on the same function because — as Lecture 6 will prove — this function is unique. The convergence equation has appeared in six fields over 150 years; TF biology is one of them.

8. What tSFM Does Differently

If the PWM is already the convergence equation for TF-DNA binding, why build tSFM at all?

The PWM limitation. A PWM is specific to one TF. To predict binding for TF_X, you need TF_X's PWM, which requires experimental data (SELEX, ChIP-seq) for that specific TF. For TFs without measured PWMs — and most of the ~1,600 human TFs lack high-quality PWMs — the standard PWM approach cannot help.

What tSFM adds. Rather than a separate PWM per TF, tSFM learns a single model that generalizes across TFs. Given a TF protein sequence, tSFM predicts what DNA motifs it will bind, without having seen that TF during training. This is cross-TF transfer — learning the relationship between TF protein structure and DNA binding preferences.

Architecturally:

- PWM: hand-crafted TF features (the PWM matrix) from experiments on that TF
- tSFM: learned TF features (ESM-2 embedding) applicable to any TF

The attention operation is the same. The difference is whether the agent features are hand-crafted per TF or learned across all TFs simultaneously.
The vibe-coding evidence. tSFM was built as one of the six SFMs in Reddy (2026). It learns TF→DNA mapping at the **structural-class level** (zinc finger, homeodomain, bHLH) for novel TFs held out from training. The temporal split result — 38.7% top-1 family retrieval, 71.3% top-5 on 178 novel post-2022 TFs — shows tSFM generalizes across TFs where a PWM-per-TF approach cannot operate at all. The 15–37% in-distribution R@1 is modest compared to miR-SFM's 98.4%, reflecting the challenge of TF→DBD-family discrimination (harder than position-in-motif discrimination). But the principle holds: one architecture, learned features, cross-TF generalization via the convergence equation.

9. Higher-Order PWM Refinements

Like the SantaLucia model, the standard PWM is a first-order approximation. Known deviations:
Dinucleotide effects (Stormo & Zhao 2010). Some TFs show dependencies between adjacent positions — the probability of base b at position p depends on the base at position $p+1$. The dinucleotide-corrected scoring function:

$$S_{\text{2nd-order}}(d)=\sum_p W^{(1)}_{p,d_p}+\sum_p W^{(2)}_{p,d_p,d_{p+1}}$$

The second-order term is a sum of position-specific, sequence-specific corrections. **In transformer terms, this maps to relative positional encoding** (Shaw et al., 2018) — an attention bias that depends on the relative position of query and key, which is exactly what a second-order PWM implements.
Long-range interactions. Some TFs (e.g., ETS family) show contacts between non-adjacent positions. In transformer terms, this is **full self-attention within the DNA sequence**, allowing any position to modify the effect of any other.
DNA shape features. TFs recognize not just base identity but also DNA shape parameters (minor groove width, roll, twist, propeller twist). Mapping to transformers: **additional input features** in the DNA encoder — embeddings that include shape in addition to sequence.
Every biological refinement maps to a specific architecture choice that transformers already have tools for. This is not a coincidence — it is the convergence equation in action.

10. Summary

Concept	Result	Significance
PWM definition	$L\times 4$ matrix of log-odds	Hand-crafted TF features from experiments
Scoring operation	$S=\sum_p W_{p,d_p}$	The field-standard TF scoring (since 1984)
One-hot DNA	$X\in\{0,1\}^{L\times 4}$	Canonical categorical encoding
Dot product form	$S=\text{tr}(W^T X)=w\cdot x$	PWM IS dot product
Boltzmann selection	$P(j)=\exp(S_j)/Z$	PWM + Boltzmann = convergence equation
Transformer identity	$w\leftrightarrow q$, $x\leftrightarrow k$, $\tau\leftrightarrow\tau$	PWM scoring IS attention
tSFM contribution	Learned q for any TF	Generalization across TFs

11. Checkpoint Questions

Q1. For the PWM in §5 (preferring ACGT), compute the score for every possible mutant at position 2 (i.e., ACGT, AAGT, ACGT→AGGT, ATGT). Which mutant drops the score most, and how does this map to the concept of "information content" at each position?
Answer: The position-2 PWM row is $(W_{2,A},W_{2,C},W_{2,G},W_{2,T})=(-1,2,0,-1)$. Scoring each mutant:

- ACGT (wild-type): $S = 2 + 2 + 2 + 2 = 8$
- AAGT: $S = 2 + (-1) + 2 + 2 = 5$
- AGGT: $S = 2 + 0 + 2 + 2 = 6$
- ATGT: $S = 2 + (-1) + 2 + 2 = 5$

A→A or A→T drops the score by 3 (most severe), A→G drops by 2, C stays at the max. The mutants that cause the largest drops are “A” and “T” — because C is strongly preferred at position 2 and its “opposites” (A, T) are most depleted. In information-theoretic terms, the “information content” at position 2 is high (roughly 2 bits — close to the maximum of $\log_2 4 = 2$), meaning position 2 is very specific and mutations away from C are highly penalized. Position 2 in this PWM is an “anchor” position for the motif.

Q2. A real transcription factor’s PWM has most positions at low information content (roughly 0.3–0.7 bits) and 1–2 “anchor” positions at high information content (1.5–2.0 bits). Explain this structure physically and map it to transformer attention.

Answer: Physically, a TF contacts the DNA major groove primarily through direct H-bond and van der Waals contacts at a few specific positions, with other positions making more promiscuous contacts that tolerate multiple bases. The anchor positions are where the TF “reads” specific chemical groups; the flanking positions are less discriminating. In transformer language, the PWM is a vector of attention logits — the anchor positions contribute large magnitude scores (high weight in the dot product), while the flexible positions contribute small magnitude scores (low weight). This is the same behavior as a learned attention distribution that focuses strongly on a few important tokens and diffusely on the rest — the pattern that interpretability work has observed in trained transformer attention heads.

Q3. Why doesn’t the standard PWM framework work for antibody-antigen binding?

Answer: Several reasons: 1. **No fixed motif length.** Antibody binding sites (epitopes) vary in length and shape; antibody CDR loops vary in length and conformation. There’s no fixed L. 2. **Non-sequential contacts.** Antibody-antigen contacts are 3D — residues at distant sequence positions come into contact through folding. Sequence-position-based features can’t capture which residues are actually proximate. 3. **No positional independence.** Antibody-antigen interfaces show strong non-additive effects — a mutation at one interface residue affects the role of other interface residues. The nearest-neighbor approximation fails. 4. **Large combinatorial space.** 20 amino acids × ~150 residues on each side gives a combinatorial space far too large for tabulation.

This is why antibody-antigen binding is the prototype of emergent bilinearity (see WB3). The dot product exists in embedding space, not sequence space. The encoder must learn it.

Q4. A skeptic says: “You’re cherry-picking — you claim the PWM IS transformer attention, but transformer attention has learned queries and keys. The PWM is hand-crafted. These are different things.”

Answer: The skeptic confuses two separate claims. Claim 1: “The PWM scoring operation is mathematically a dot product.” This is a mathematical identity — true regardless of how the PWM’s numerical values were obtained. Claim 2: “PWMs are hand-crafted from data, while transformer queries are learned via backpropagation.” This is true but is a claim about **how the parameters are obtained**, not about **what the operation computes**. Both PWM scoring and transformer attention compute dot products between agent features and target features, followed by softmax. Whether the features are obtained by tabulation (PWM), by an ML algorithm with gradient descent (transformer), or by physical measurement (SantaLucia parameters) does not change the operation. The convergence equation concerns the operation, not the source of the parameters. The distinction between “hand-crafted” and “learned” maps directly to “given” vs. “emergent” bilinearity — both yield the same mathematical form.

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12. Connection to Future Lectures

- **Lecture 5 WB3:** Emergent bilinearity — what happens when the dot product exists only in an embedding space that has to be learned.

- **Lecture 6:** Proves that the softmax form applied to these dot products is the unique selection function satisfying five axioms of competitive selection.
- **Lecture 7:** Derives the complete SFM architecture from the convergence equation. For given-bilinearity domains like TF-DNA, the architecture implements the PWM-equivalent operations with learned TF embeddings.
- **Lectures 9–11:** Students building tSFM will work with ESM-2 (TF encoder) and DNABERT-2 (DNA encoder). The dot-product scoring shown in this guide is exactly what tSFM computes at inference time.
- **Lecture 10 (advanced):** The affinity calibration framework $\Delta G = ws + b$ (future work) shows that the contrastive score s is proportional to binding free energy. For given-bilinearity systems like tSFM, the calibration is tight because the learned features mirror the physical parameters — the SFM has "rediscovered" the PWM.